

1. Objective and implemented architecture

The assignment asks for dimensioned, stitched property plans from phone photos, video and LiDAR, with openings, damage, confidence and independently measured accuracy. The intended users are property restoration teams. This submission is a local Python command-line prototype using a stock-phone capture route.

The implemented sequence is capture intake and provenance, sensor-specific reconstruction, supported structural geometry, optional verified stitching, room/surface assessment, openings and damage, uncertainty/contract checks, and review exports. Missing support produces unavailable measurements or partial results. Reporting success and geometric acceptance are separate decisions.



`workflow.run_capture` orchestrates intake, reconstruct and assessment writing. `ingest` preserves frame identity and metadata; `sfm/rgbd` supply reconstruction; `layout` handles finite walls and local floor/ceiling support. `stitching` is an explicit operation, not an automatic guarantee. `assessment`, `openings`, `damage` and `uncertainty` produce the internal contract. `assignment_gates`, `repeatability` and `consumer evaluators` compare outputs with external references.

Outputs include JSON and offline HTML; supported geometry adds SVG, DXF, CSV, cloud/trajectory and diagnostic artifacts. The published assignment schema was not supplied, so internal schema validation is provisional.

Conservative acceptance preserves observed geometry and withholds unsupported ceilings, scale and connections. CPU execution and explicit source/model hashes favor inspection and reproducibility; limited observations and optional native dependencies constrain completeness and startup time.

2. Input tiers, calibration and uncertainty

Tier	Implemented path	Current boundary
Photos	Room folders, EXIF/HEIC handling, SIFT/COLMAP sparse cameras and tracks	Strict RGB scale/property placement unresolved; no measured all-room acceptance
Video	Original MOV/MP4, timestamped frame sampling, same production RGB path	Low-detail transition remains unresolved; investigation closed as inconclusive
LiDAR	Stray depth/confidence, per-frame intrinsics and odometry; calibrated backprojection/fusion	Real supplied data exercised; partial rooms and missing local floor/ceiling support

SIFT remains production. Learned metric-depth inference and XFeat/LightGlue, fixed-intrinsics and mapping ablations remain experimental. Sensor poses were withheld from the RGB transition reconstruction and used only in post-hoc audits. No experimental connectivity gain is promoted to a production accuracy claim.

The supplied three cases contain RGB video, depth/confidence and camera/IMU/ odometry CSVs. They exercise calibration extraction, synchronization, fusion, support and failure behavior. They do not supply certified room dimensions, physical damage labels, independent repeats or qualifying consumer exports.

Metric LiDAR scale is sensor-derived. Strict RGB cannot obtain absolute scale from arbitrary photos alone; measured-reference research controls are separate. Distortion/aspect-ratio checks and producer-bound calibration prevent silent mixing of incompatible evidence. A camera matrix is a projection calibration, not proof of physical measurement accuracy.

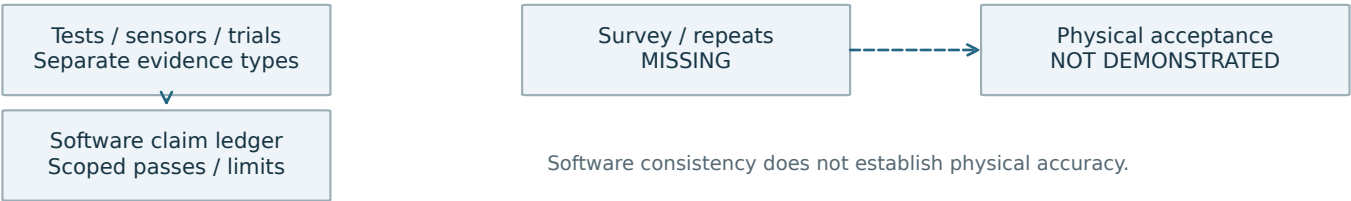
Depth-backed verified registration and pose-graph/loop mechanisms exist, but an effective physical drift correction and qualifying footprint on/off benchmark have not been demonstrated. Interval fitting/auditing is property-grouped and producer-bound; adequate independent calibration properties are absent.

The unquantified error budget includes depth bias, intrinsic/distortion error, pose drift, plane fitting, boundary/jamb localization and missing observations. No per-component centimetre allocation is supported by a physical survey. Support, reprojection error and field accuracy remain different quantities. Base iPhone hardware supports RGB; depth capture requires compatible Pro hardware. The novice/cold-device Route 2 rehearsal and timed clean installation are pending.

3. Validation evidence and benchmark results

Published evidence has SHA-256 and exact JSON field paths; original-byte provenance is recorded separately. Controlled tests establish software invariants. Supplied-sensor audits establish internal consistency; they do not replace laser/tape truth.

Claim /metric	Recorded result	Status /scope
Software regression, batch032	390 tests pass; 43.22 s	PASS, historical software run
Fixed ceiling-layout replay	15/15 exact checks	PASS, same saved artifact
Native sensor fusion	218,873 samples; 202,477 points, bit-identical arrays	PASS, internal consistency
Floor-only containment	25/176 camera samples	PARTIAL, not floor-area accuracy
Ceiling-scan containment	295/300 camera samples	PARTIAL, inferred geometry included
room_2 local floor	148 points /43 cells; 14.1246% support	PARTIAL, below unchanged 25% guard
Independent dimensions/repeats	Required truth absent	NOT DEMONSTRATED

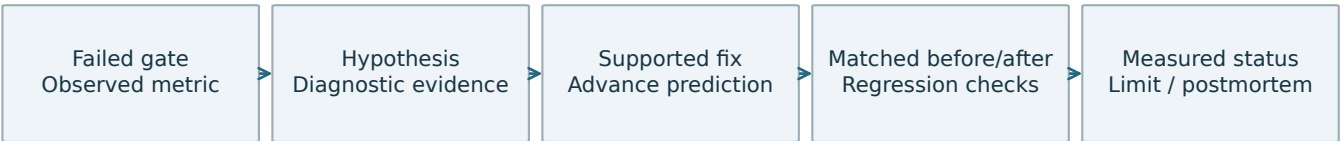


The floor-source audit found 29.2154% diagnostic occupancy from accepted pixels in the same 300 selected frames, versus 14.1246% after stride-8 sampling. All-source raw-pose support reached 44.0880%. Fusion lost zero local occupied cells. This identifies sampling/frame-selection loss, not a proven extraction defect or a validated dense-floor fix. Production geometry/guards are unchanged.

The 0.702677 m discrepancy is a floor-plan footprint notch, not a ceiling-height shift: area changed 10.477367 to 9.945720 m²; 0.531648 m² was removed. Height was unavailable before and after. Crossing rays and exact fusion do not independently identify the physical perimeter. Both interpretations remain unvalidated.

Recorded audit times exclude capture, transfer and installation. No qualifying same-property three-tier physical benchmark, consumer score or cold walk-in runtime is available. The compliance matrix preserves each unmet gate.

4. Fix loops, rejected approaches and engineering judgment



Unsupported causality → diagnostic / inconclusive; retain production.

Completed engineering milestone	Evidence /result	Remaining limit
COLMAP/OpenCV pixel-center correction, 9d75ab6	Remove analytical 0.5 px sampling offset; identity/resized-image regression	No measured wall-error improvement claimed
Finite wall/seed support, 061625b	Final bounded step: 67 to 69 fragments; prior cells preserved	Floor containment stays 25/176; completeness unresolved
Registered damage evidence, 8aaca9f	Reject unregistered/inconsistent surface observations; tested guards	No independently scored two-class field validation

These are shipped software milestones. Their prospective physical worst-gate prediction was not recorded. They cannot retrospectively satisfy the scored Fix Loop, which requires a measured failing gate, advance prediction, shipped fix and reproducible matched before/after evidence.

The matched 32-view DISK/LightGlue trial produced 167 verified pairs and one graph component, but separate 15/11-view models with 1,931/1,404 points and 1.449190/1.337805 px residuals. No joint target-group model formed. Production adoption was rejected. The retained SIFT control had 68 pairs and an 11-view model with 1,012 points and 0.636201 px residual; this is also incomplete.

Experimental XFeat mapping-only fixed intrinsics registered 32 views and 6,187 points at 1.480503 px residual, but later relative orientation disagreed with odometry by approximately 6.520814°. Final frame31 replay did not reproduce the trusted snapshot; causality remained unresolved. The registration investigation is closed. No production solver defect or physical tolerance was established.

Denser LiDAR sampling increased containment to 79/176 but moved accepted geometry, so it was rejected. Floor/ceiling source audits justified retaining the guard and unavailable height. Diagnostic restraint protects defensibility; negative results are retained rather than marketed as fixes.

5. Reproduction and final assessment status

The submitted Git tree contains the frozen implementation and required tests/ tooling. The separate package contains clean tracked source, portable evidence, supplied raw data and integrity manifests. Private document backups, environments and external model assets are excluded; original-byte provenance is retained.

Setup uses scripts/bootstrap_windows.ps1; run CLI help and python -m pytest -q from the source root. Final QA's documented LiDAR command wrote JSON/HTML/plan artifacts but exited 1: 300 frames, 202,477 points, 12 cell hypotheses, zero accepted ceiling heights and zero connections. Capture-command runtime was 221.196 s; reconstruction-ledger runtime was 69.057 s, excluding install/capture/transfer. This is partial execution evidence. Historical native producer pins, omitted experimental assets and physical truth limit every-number reproduction.

Assessment area	Final status	Missing evidence /impact
Full property geometry and strict RGB metric plan	PARTIAL	Supplied floor completeness, adjacency and RGB scale unresolved
Openings, ceiling heights, damage and concealed scope	PARTIAL	Guarded paths exist; missing local height and independent field scoring
Opening <=2 cm /85%; ceiling <=1.5 cm; repeats	NOT DEMONSTRATED	Detection-aware laser/tape truth and separate repeat captures
Photo +/-8%, video +/-3%; interval calibration	NOT DEMONSTRATED	Same-property tiers and independent calibration/audit properties
Consumer >=70% beat/tie; mandatory scored Fix Loop	NOT DEMONSTRATED	Original app exports; measured benchmark and prospective declaration
Boundary and registration causality	INCONCLUSIVE	Physical perimeter truth /reproducible causal solver evidence absent
Cold walk-in and setup time	NOT DEMONSTRATED	Unseen phone/tier rehearsal and timed fresh-machine live output

Earlier Round 1 LiDAR tolerances and the official output schema were not included; no substitute is invented. Real observations of at least three rooms plus a connector, furnished/staged damage, same rooms at every tier, repeated capture, laser/tape survey and consumer exports remain necessary for acceptance.

Final conclusion: the current implementation is packaged for honest engineering evaluation, with traceable production fixes, experimental results and material limitations. Software consistency is demonstrated on stated inputs. Independent centimetre-level accuracy, full completeness and assessment acceptance are not.

More detail: architecture, validation, compliance, Fix Loop, limitations and reproduction. Exact source fields and hashes are indexed in the evidence manifest.